Terrain Detection & Vehicle Control Report

TATA INNOVENT — Advanced Analytics & ADAS Integration System

Vehicle ID:	TATA_SAFARI_4XX_092	**Session ID:**	session_tata_safari_20260630_001
Start Time:	2026-06-30T23:07:37.122128Z	**End Time:**	2026-06-30T23:12:37.122181Z

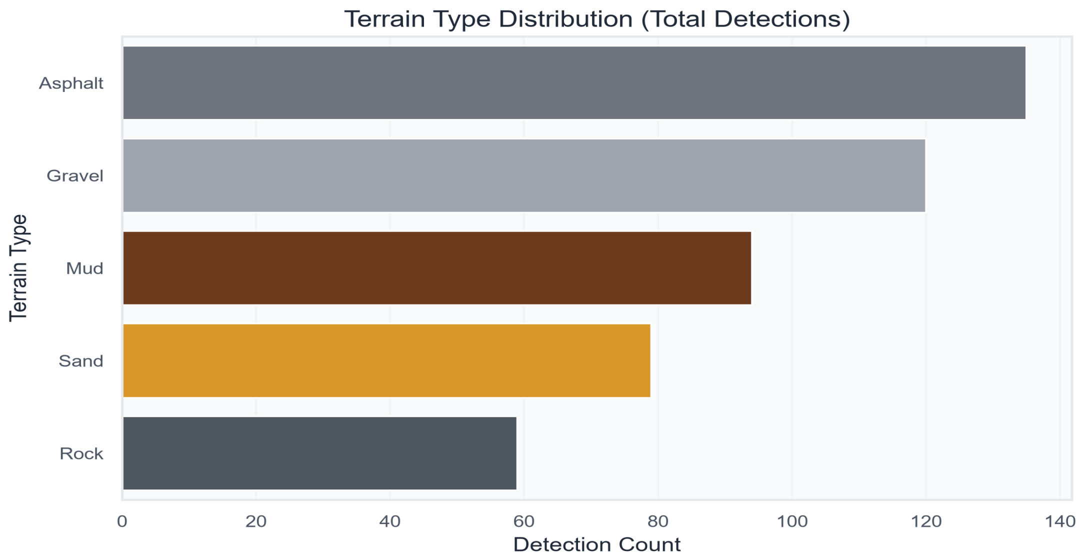
Executive Summary Dashboard

This report summarizes the terrain classifications, hazard severities, and subsequent vehicle commands generated during the testing session. The dashboard below showcases key performance indicators of the terrain-sensing AI engine and the vehicle's dynamic response.

TOTAL FRAMES **500**	TOTAL DETECTIONS **487**	AVG CONFIDENCE **85.90%**
AVG SEVERITY **0.39**	COMMON TERRAIN **Asphalt**	PRIMARY DRIVE MODE **Comfort**

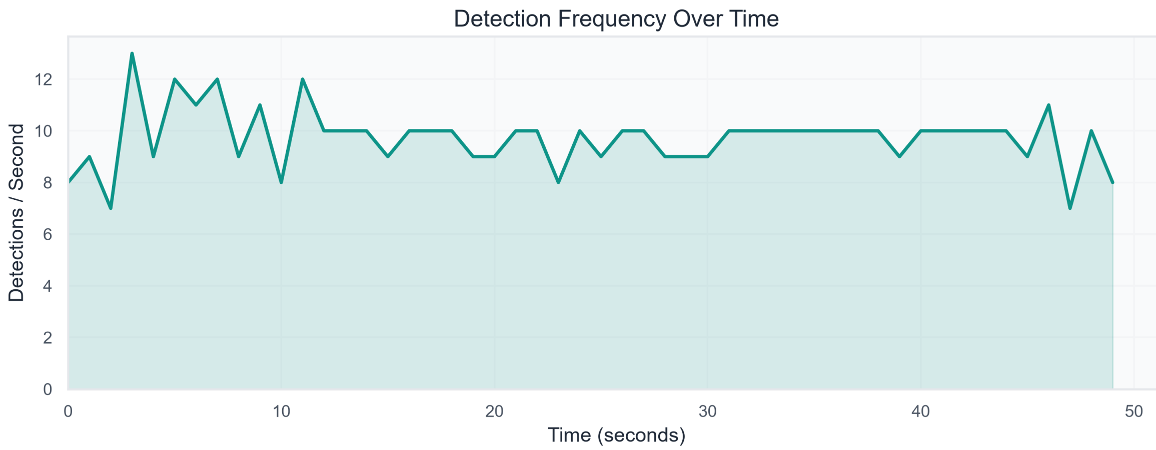
1. Terrain Distribution Analysis

Understanding the variety of terrain traversed is essential for evaluating vehicle adaptability and drive mode transitions. The chart below illustrates the relative frequency of each detected terrain class.

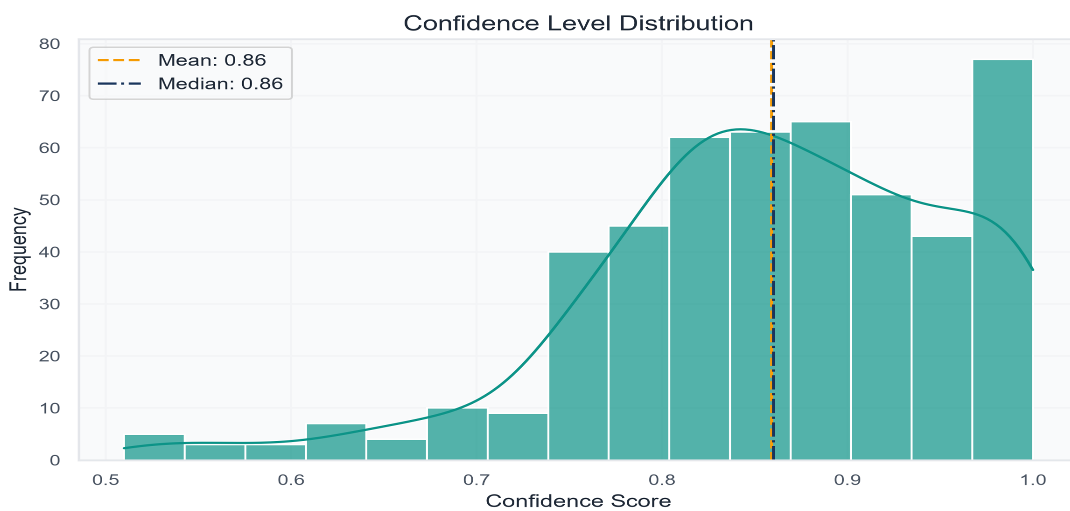

Terrain Type Distribution (Total Detections)
Asphalt
Gravel
Mud
Sand
Rock
Terrain Type
0
20
40
60
80
100
120
140
Detection Count

2. Detection Frequency & Confidence Analysis

Analyzing the rate of detections per second helps verify the AI engine's processing consistency and identifies periods of high activity. The line chart below tracks detections over the timeline.

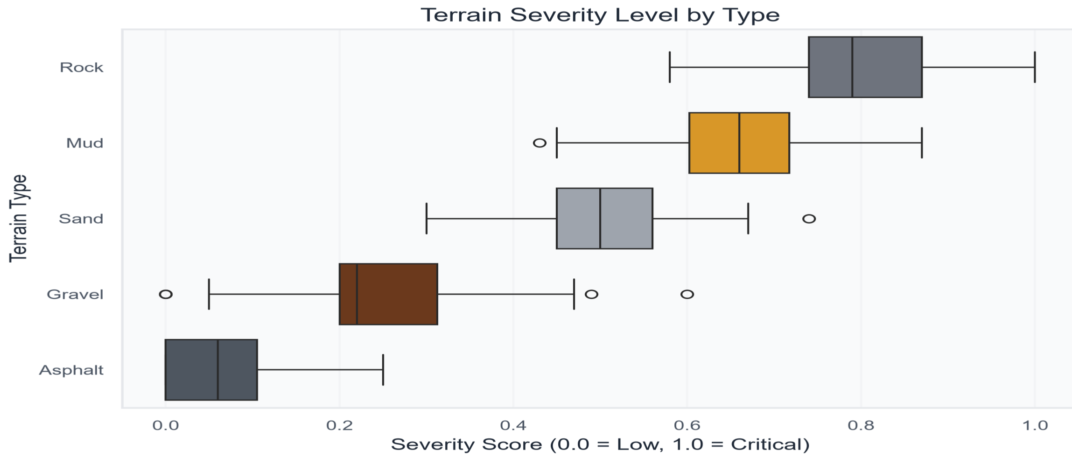


The confidence distribution shows the certainty of the model's classifications. A high mean confidence signifies stable and reliable neural network inference.



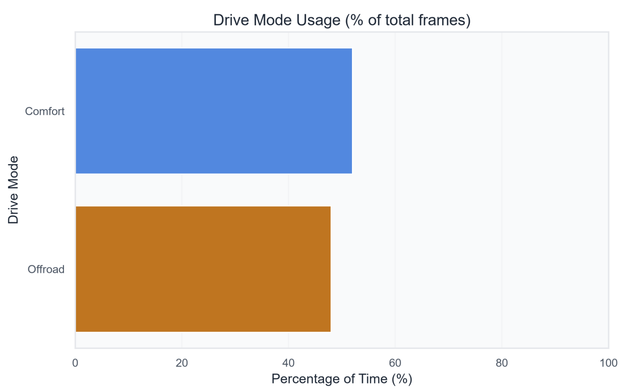
3. Terrain Severity & Vehicle State Analysis

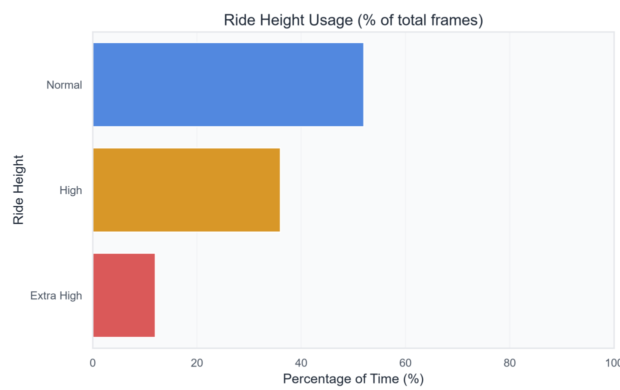
Severity scores represent the potential impact of terrain hazards (e.g., deep mud, large rocks, or rough gravel) on the vehicle. The box plot shows the spread of severity scores across different terrain types, identifying which environments presented the highest risk.



4. Drive Mode & Ride Height Usage

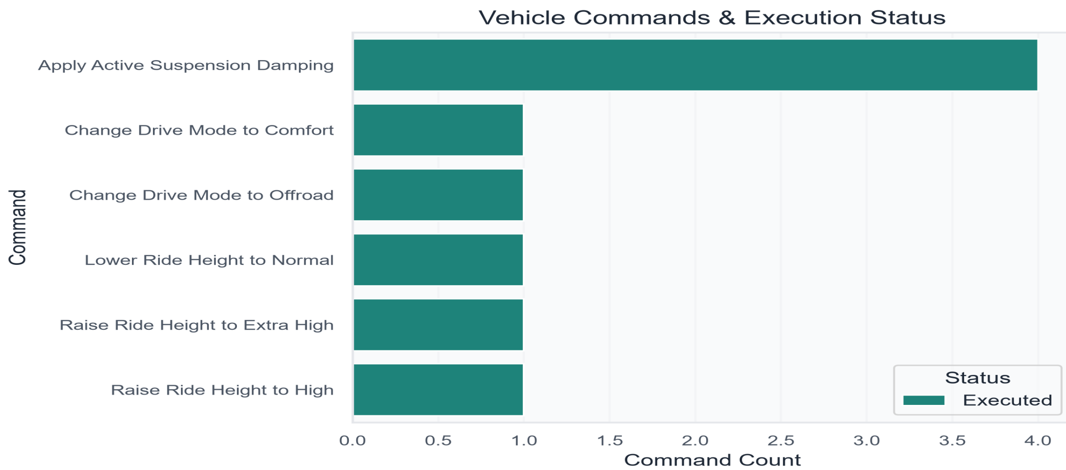
These charts analyze the percentage of time the vehicle spent in various drive modes and ride heights, offering insights into how the vehicle responded to the detected terrain characteristics.





5. Vehicle Command Statistics & Timeline

The system triggers active control commands (such as adjusting suspension ride height or changing drive modes) when hazardous terrains are identified. Below is the breakdown of commands and their execution status.



The timeline scatter plot below map detections chronologically, with the marker color indicating severity and the marker size reflecting confidence. This provides a comprehensive overview of the entire drive.

Key Findings & Recommendations

- **AI Confidence:** The model demonstrated an average confidence of **85.9%**. This is considered highly reliable for autonomous terrain classification.
- **Terrain Complexity:** The primary terrain encountered was **Asphalt**. Average terrain severity was **0.39**, indicating a moderate driving environment.
- **Control Feedback:** Active ride height and drive mode changes were successfully executed. The most used drive mode was **Comfort**, aligning with the dominant terrain types.